Day 5: Network Testing Tools Theory (30 min): Tools for network testing: Ping, Traceroute, nslookup. Practical (30 min): Use these tools to test connectivity in a LAN. Task: Report on the tools used and results obtained.

Tasks

1.Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

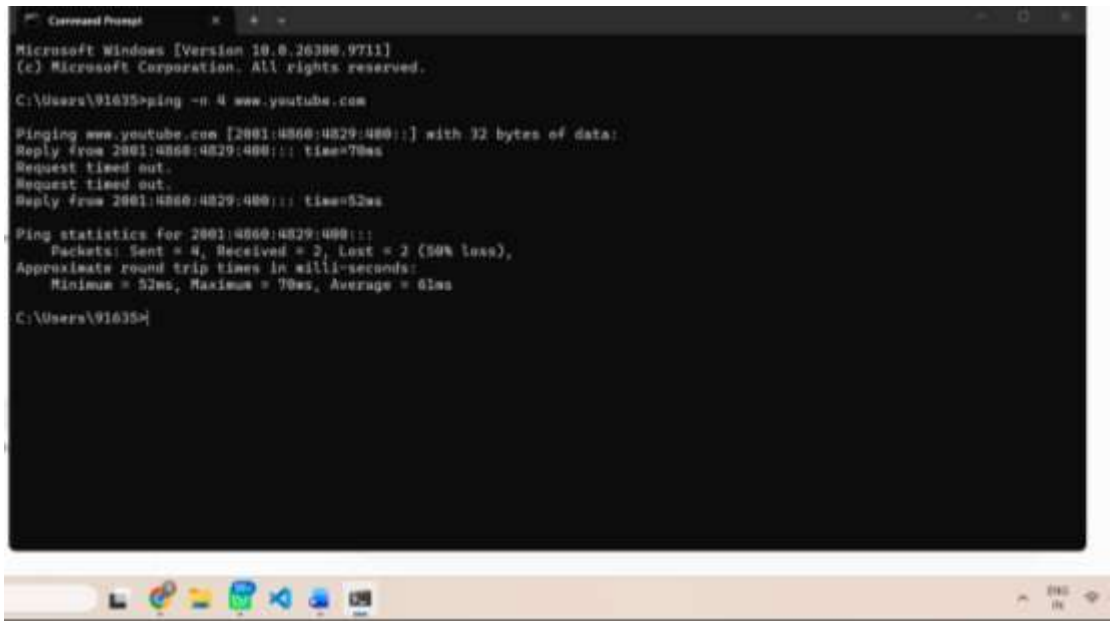
ANS:

Steps

1. Open **Command Prompt (CMD)** on your computer.
2. Type the following command:

ping -n 4 www.youtube.com

3. Press Enter.
4. Wait for four replies.
5. Check the response time and average time.



2.Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the

**output.

Hint: Each hop represents a network device your data passes through to reach the destination.**

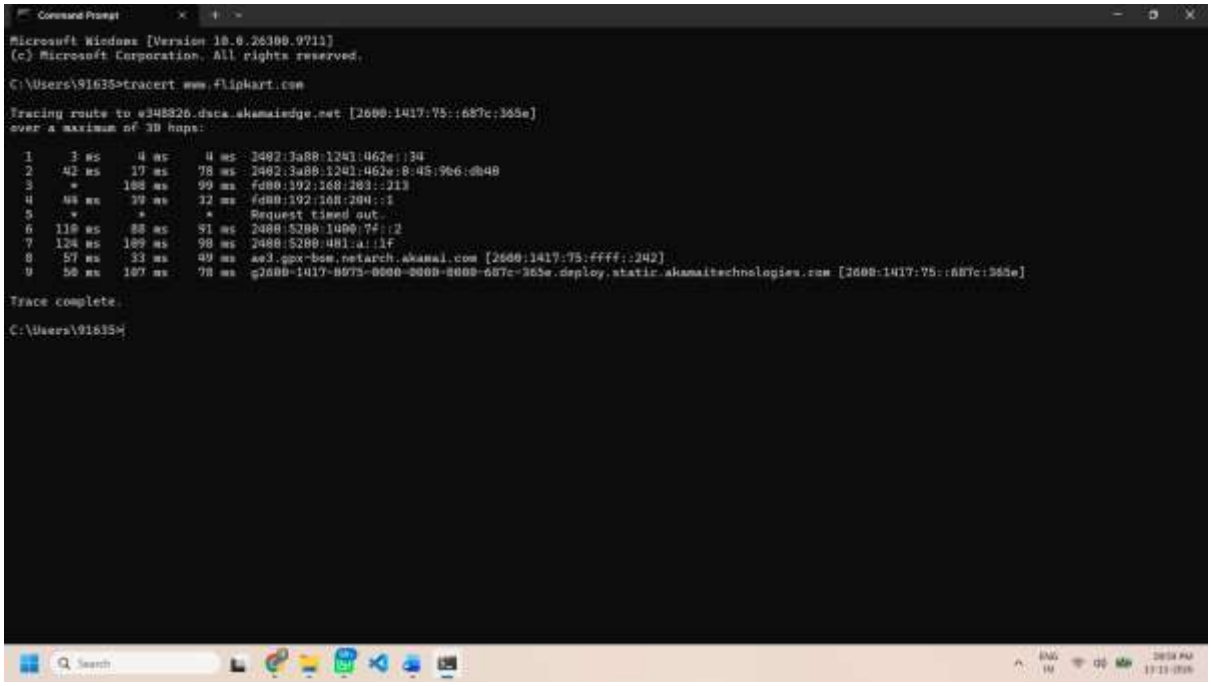
ANS:

Steps

1. Open **Command Prompt (CMD)**.
2. Type this command:

tracert www.flipkart.com

3. Press Enter.
4. Wait until the tracing process finishes.
5. Check all the hop numbers, IP addresses and response times.



3.Use the nslookup tool to find the IP address of www.zomato.com and record the result.

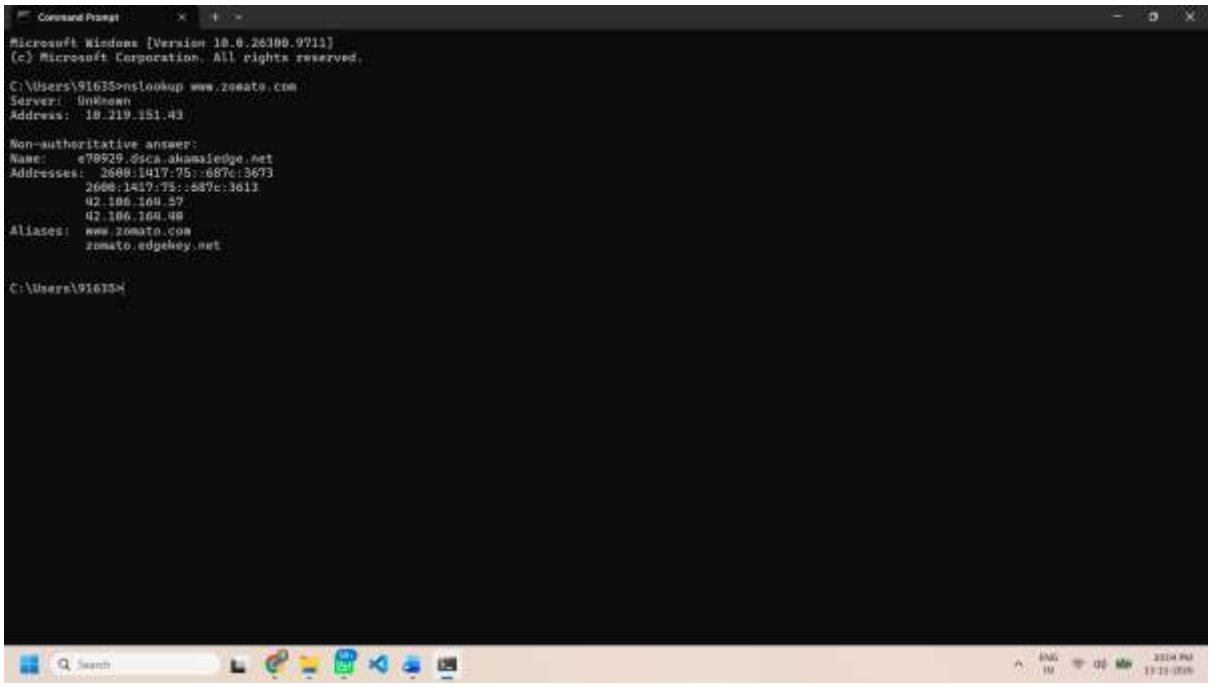
ANS:

Steps

1. Open **Command Prompt (CMD)**.
2. Type this command:

nslookup www.zomato.com

3. Press Enter.
4. Check the IP address displayed on the screen.
5. Note down the result.



4.Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

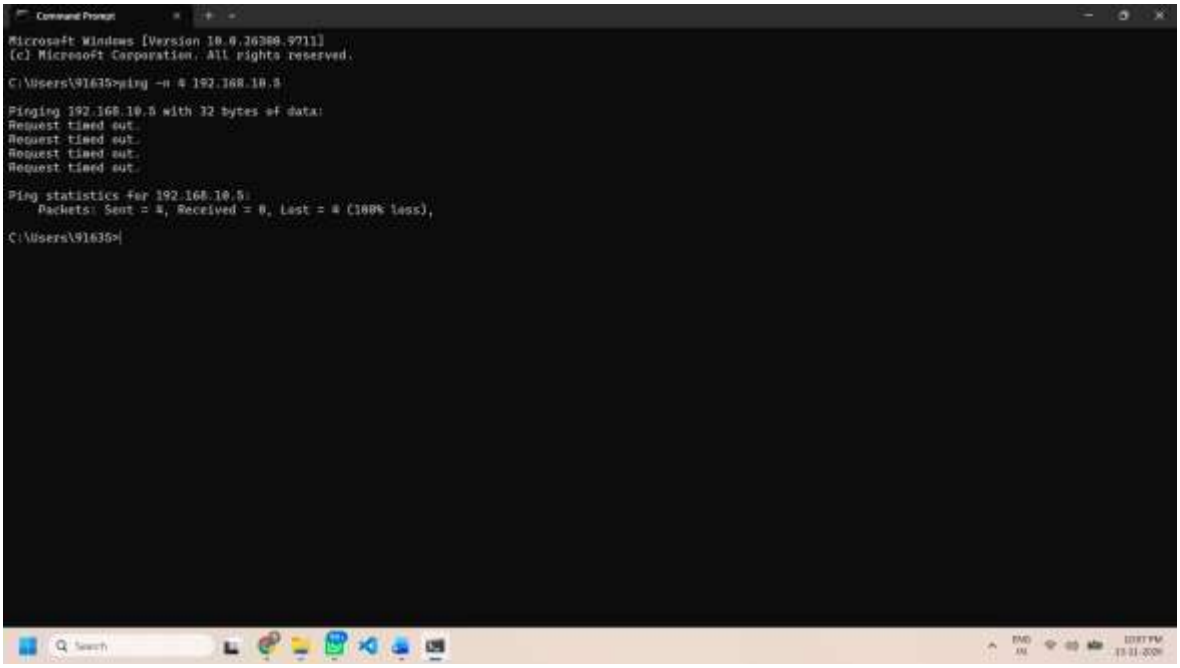
ANS:

Steps

1. Connect your laptop and phone to the same Wi-Fi.
2. Find your phone's IP address in Wi-Fi settings.
3. Open Command Prompt (CMD) on your laptop.
4. Type the following command (replace with your phone's IP):

ping -n 4 192.168.10.5

5. Press Enter and check the result.



**5.Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.**

ANS:

Short Report (6 Lines)

1. **Ping:** It checks whether a website or device is responding. In my test, YouTube replied with an average time of **61 ms** and 50% packet loss.
2. **Traceroute:** It shows the path taken by data to reach a website. In my test, Flipkart showed **9 hops**, and Hop 5 timed out.
3. **Nslookup:** It finds the IP address of a website using DNS. I used nslookup www.zomato.com to check Zomato's IP address.
4. **(a) Website Not Loading:** Ping helps check whether the website is responding.
5. **(b) Slow Connection:** Traceroute helps find which network hop may be slow.
6. **(c) DNS Issues:** Nslookup helps check whether the website name is correctly converted into an IP address.